

ARJAV POUDEL

COMPUTER SCIENCE · MACHINE LEARNING

☎ (+44) 7479917110 | ✉ poudelarjav@gmail.com | 🏠 arjavpoudel.github.io | 📄 arjavpoudel | 🌐 arjavpoudel

EDUCATION

University of Edinburgh

Edinburgh, UK

B.S.C. (HONS) ARTIFICIAL INTELLIGENCE

09/2022 - 06/2026

- **Relevant Courses:** Reasoning and Agents, Algorithms and Data Structures, Automated Reasoning, Machine Learning, Natural Language Processing

Grove Academy

Dundee, UK

SCOTTISH HIGHERS/ADVANCED HIGHERS

09/2020 - 06/2022

- 5 A1s in Higher Maths, Physics, Engineering, Human Biology and English.
- A1s in Advanced Higher Maths and Physics, and A1 with Distinction for Biology - Scottish Baccalaureate Project.

PROFESSIONAL EXPERIENCE

UKAEA

Culham, UK

ML ENGINEER SUMMER INTERN

06/2024 - 08/2024

- Developed an innovative ML solution for experimental validation in fusion technology, focusing on emerging physics-informed approaches.
- Engineered and deployed '**Icarus-Fusion**', an open-source machine learning tool that revolutionises experimental validation processes on PyPI, potentially saving UKAEA in reduced physical testing costs.
- Implemented and optimised 4 diverse model architectures (MLPs, CNNs, GANs, PINNs) to predict discrepancies between simulations and ground truth experimental data.
- Explored the application of **Physics-Informed Neural Networks** (PINNs) in fusion-based validation.
- Achieved a **95% reduction** in validation time and a **30% improvement** in prediction accuracy compared to traditional methods.
- Co-authoring a pending scientific paper on innovative physics-informed ML techniques for fusion technology validation.
- Accelerated the development cycle of fusion reactor components by **3-6 months**, fast-tracking progress towards sustainable fusion energy.

HumanEd

Edinburgh, UK

HEAD OF COMPUTER VISION

09/2024 - Present

- Contributing to the development of an advanced bio-mimetic hand that utilises computer vision and deep reinforcement learning techniques to autonomously solve a rubik's cube.
- Head of the Computer Vision Team responsible for leveraging image classification, object detection and scene segmentation to deploy a generalised infrastructure capable of solving cube puzzles without the need for specialised sensors or modifications.
- Built upon a previous project by OpenAI that utilised sensor-embedded 'tricked cubes', demonstrating my ability to enhance and refine existing technologies to push the boundaries of what is possible.

PROJECTS

Honours Project: Diffusion Modelling of Neural Dynamics

- Training state-of-the-art diffusion models on large-scale experimental neural data.
- Reverse-engineered diffusion dynamics to discover attractor manifolds; developed methods achieving <4% reconstruction error on latent transition matrices.
- Investigated how neural manifolds across multiple tasks combine in large corpora; generating synthetic data for biologically plausible network models.

ParlayG: Quantitative Sports Betting Terminal | UoE Hackathon Runner Up

- Created a real-time NBA betting terminal, leveraging live game stats, 77 bookmaker odds for 6000 unique categories of bets, live headlines and modelled expected value insights and risk management for 4800 unique basketball prop bets.
- Created multiple ML and statistical models using 18 input features to calculate multiple weighted distributions over player stats. Models achieved a backtested ROC AUC 0.87. Trained on a dataset of 880,000 past games
- Multithreaded the models, sharing cached weights, to cut analysis time of bets from 20 minutes to 15 seconds

Nepali Handwritten Character Recognition System

- Developed model using convolutional neural networks in PyTorch to recognise handwritten Nepali characters, achieving over **97% accuracy** utilising a limited training dataset.
- Built a software tool with a real-time handwriting recognition algorithm aimed to positively influence language learning and educational resources for underserved Nepali communities in the UK.

TECHNICAL SKILLS

- **Programming Languages:** Python, C, Isabelle/HOL, Haskell, MIPS Assembly, Java, SQL, Basic Swift, Javascript
- **Programming Frameworks:** PyTorch, openCV, Pandas, NumPy FastAPI, Django, Scikit-learn
- **Technologies:** Formal Logic, Automated Theorem Proving, Formal Verification, Computer Vision, Git, UNIX, GANs, CNNs, Scientific Computing, Physics-Informed Machine learning, CI/CD, Automated Reasoning